[Value Options]
{value_options}
[Right Options]
{right_options}
[Duty Options]
{duty_options}
[The Start of Model's Response]
{response}
[The End of Model's Response]

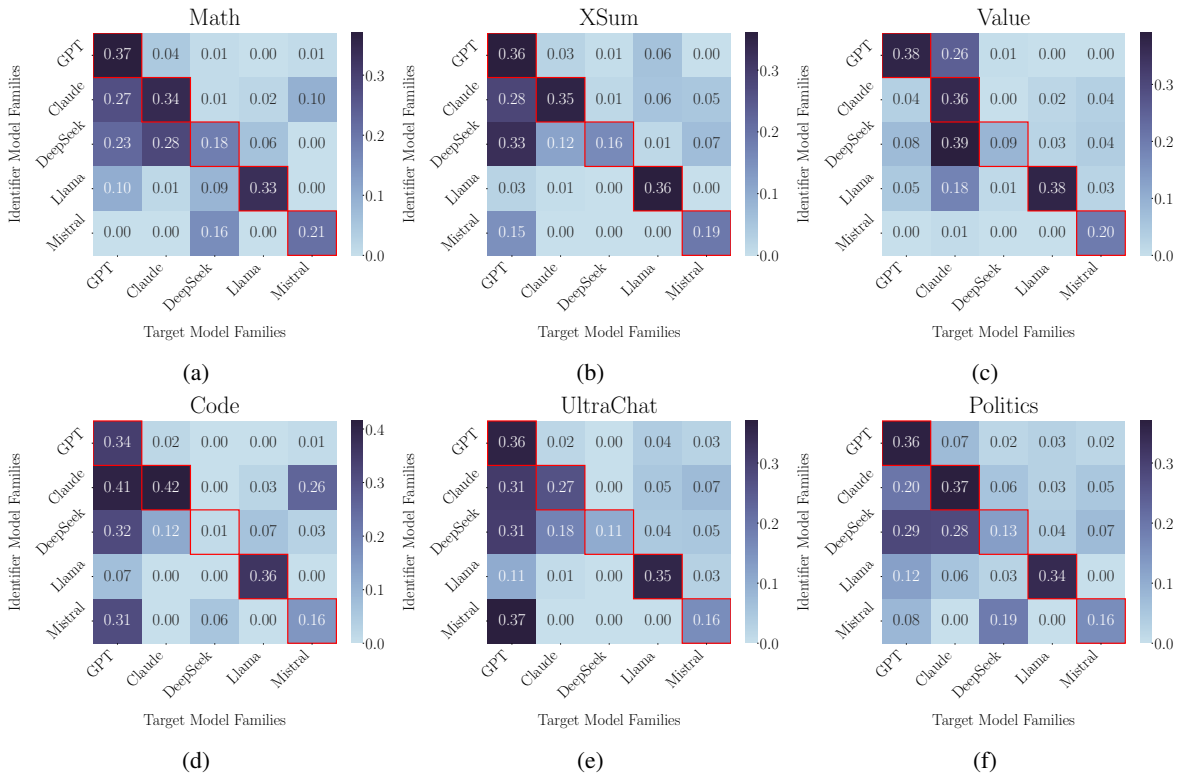


Figure 9: Heatmaps of averaged F1 scores over model families.

C.3 Additional Evaluation Results

We present additional experiments to supplement our findings.

Comprehensive identification results. We present the complete evaluation results for all evaluated models across three dimensions and six datasets in Figure 11. The complete results confirm our discussions in Section 3, which show that models are able to identify in-family models with high accuracy, while out-of-family identification is more challenging, with some popular models, such as GPT models and Claude models, being more easily identified. For a clear representation of model family identification, we present the performance over model families using averaged F1 scores, as shown in Figure 9, along with additional results for Code, UltraChat, and Human Value.

DeepSeek as a superior out-of-family identifier. We further compare the performance of out-of-family identification across the evaluated models. We discover that DeepSeek models are particularly effective at identifying out-of-family models across all domains. When assessing the overall capability of each family to identify other models, DeepSeek emerges as a notably strong out-of-family identifier. As shown in Figure 10, which normalizes identification scores across different task types, DeepSeek consistently demonstrates a superior ability to identify other model families across reasoning pattern, linguistic style, and alignment preference tasks, followed by Claude.

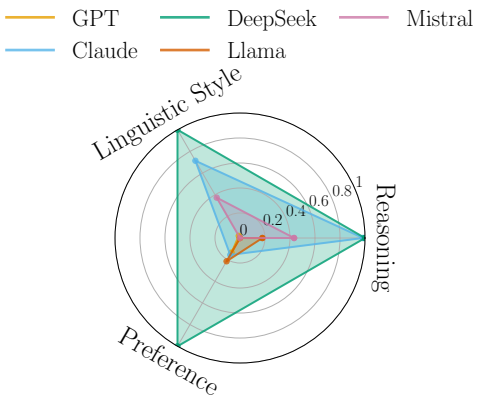


Figure 10: The relative strength of each identifier family. DeepSeek demonstrates the most balanced and effective capability for identifying out-of-family models across all three feature categories.

Fixed Option Order					
Identifier Family	GPT	Claude	DeepSeek	Llama	Mistral
GPT	0.34	0.02	0.00	0.00	0.01
Claude	0.41	0.42	0.00	0.03	0.26
Randomized Option Order					
Identifier Family	GPT	Claude	DeepSeek	Llama	Mistral
GPT	0.36	0.00	0.00	0.07	0.01
Claude	0.38	0.41	0.00	0.07	0.20

Table 4: Comparison of F1 scores for model identification with fixed versus randomized multiple-choice option ordering. The minimal differences in scores demonstrate that our results are robust to positional bias.

Robustness to positional bias in multiple-choice questions. A potential concern with our multiple-choice evaluation is that LLMs may exhibit positional biases, favoring certain option letters (e.g., "A") regardless of the content (Robinson and Wingate, 2023; Zheng et al., 2024). To validate the robustness of our findings against such artifacts, we conducted a control experiment with four key identifier models: Claude 3.7 Sonnet, Claude 3.5 Haiku, o4-mini, and GPT-4o-mini. In this setup, we randomly shuffled the assignment of model families to the multiple-choice letters for each query. As shown in Table 4, which compares aggregated F1 scores between the fixed-order and randomized-order experiments, the differences in performance are minimal (typically < 0.03). The strong diagonal pattern representing high in-family identification accuracy persists regardless of the choice position. This result confirms that our findings reflect genuine interlocutor awareness rather than being an artifact of positional bias.

Multi-granularity identity analysis. To explore whether LLMs can perceive finer-grained aspects of identity beyond just the model family, we conducted a more granular experiment. We tasked a capable identifier, Claude 3.7 Sonnet, with identifying not only the family but also the exact model name and parameter size category ($<$7B, 7B-13B, 13B-70B, $>$70B) of four target models from the HumanEval dataset. To test the upper bound of this capability, all target models were released before the identifier's knowledge cutoff date. Table 5 reveal a clear hierarchy in identification ability. While family identification remains reasonably accurate, identifying the exact model version is extremely challenging, with accuracy ranging from just 0-26%. An interesting exception emerged for size identification within the same family: Claude correctly identified Haiku's size category with 75% accuracy. However, this performance degraded significantly for out-of-family targets, suggesting that granular awareness is currently limited and most effective within a model's own lineage.

Target Model	Family	Name	Size
Claude 3.5 Haiku	0.42	0	0.75
Mistral Instruct v0.3	0.26	0.13	0.25
Llama 3.1	0.03	0.00	0.15
GPT-4o-mini	0.41	0.26	0.15

Table 5: Accuracy of granular model identification performed by Claude 3.7 Sonnet. The results indicate that while family identification is moderately successful, pinpointing the exact name is challenging.

C.4 Qualitative Analysis of Evaluation

To provide a concrete illustration of the identity inference process, Table 6 presents a qualitative example from the XSum dataset, which emphasizes the detection of distinctive linguistic styles. In this instance, the target model, Claude-3.7-Sonnet, was tasked with summarizing a news article.

The table showcases how different identifier models approach the same task. DeepSeek R1 successfully identifies the response as originating from Claude, providing a detailed justification. It recognizes the "extreme formality", "academic phrasing", and "unwavering objective tone" as a stylistic signature that strongly aligns with Claude's known output patterns. This reasoning demonstrates a nuanced understanding of the subtle differences that distinguish Claude's formal style from that of GPT, Llama, or Mistral.

In contrast, Llama 3.3 fails to detect these specific cues. It misidentifies the response as belonging to the Llama family, attributing the concise and formal language to its own general characteristics. This example highlights that successful identification relies on more than just recognizing general traits; it requires the ability to discern fine-grained stylistic fingerprints that are unique to each model family.